

HDOC Day-34

Distributive Properties of Matrices Using Variable-Length 2D Arrays in C

Problem Statement

Create a menu-driven C program using separate functions to verify the left distributive property $A \times (B + C) = (A \times B) + (A \times C)$ and the right distributive property $(A + B) \times C = (A \times C) + (B \times C)$. Use variable-length 2D integer arrays, display both sides, and state whether the property is verified.

Algorithm

1. Display the menu repeatedly until Exit is selected.
2. Choice 1: read positive m, n and p ; read $A(m \times n)$, $B(n \times p)$, $C(n \times p)$; compute $B+C$; calculate $A \times (B+C)$, $A \times B$ and $A \times C$; compare $A \times (B+C)$ with $(A \times B) + (A \times C)$; display both matrices and the result.
3. Choice 2: read positive m, n and p ; read $A(m \times n)$, $B(m \times n)$, $C(n \times p)$; compute $A+B$; calculate $(A+B) \times C$, $A \times C$ and $B \times C$; compare $(A+B) \times C$ with $(A \times C) + (B \times C)$; display both matrices and the result.
4. Choice 3: exit. For any other choice, display an invalid-choice message.
5. Whenever m, n or p is non-positive, reject it and ask again.

Test Case Table

TC	Choice	Input	Expected Result
1	1	$m=2, n=2, p=2$; $A=1\ 2\ / \ 3\ 4$; $B=5\ 6\ / \ 7\ 8$; $C=1\ 1\ / \ 1\ 1$	Both sides = $22\ 25\ / \ 50\ 57$; left distributive property verified.
2	1	$m=1, n=2, p=1$; $A=2\ 3$; $B=1\ / \ 4$; $C=5\ / \ 2$	Both sides are equal; left distributive property verified.
3	2	$m=2, n=2, p=2$; $A=1\ 2\ / \ 3\ 4$; $B=5\ 6\ / \ 7\ 8$; $C=\text{identity } (1\ 0\ / \ 0\ 1)$	Both sides = $6\ 8\ / \ 10\ 12$; right distributive property verified.
4	2	$m=1, n=2, p=1$; $A=1\ 2$; $B=3\ 4$; $C=5\ / \ 2$	Both sides are equal; right distributive property verified.
5	1	Enter $m=0$, then $m=2$; enter $n=0$, then $n=2$; $p=2$; then valid matrices	Rejects 0 values, asks again, then completes Choice 1 normally.
6	2	Enter $p=-1$, then $p=2$; then valid matrices	Rejects -1, asks again, then completes Choice 2 normally.
7	9	Invalid menu choice	Invalid choice. Please enter 1 to 3.
8	3	Exit	Exiting program.

C Program

```
#include <stdio.h>
```

```
void leftDistributive(void);  
void rightDistributive(void);
```

```
int main(void) {  
    int choice;  
    do {  
        printf("\n--- MATRIX DISTRIBUTIVE PROPERTY MENU ---\n");  
        printf("1. Verify left distributive property\n");  
        printf("2. Verify right distributive property\n");  
        printf("3. Exit\n");  
        printf("Enter your choice: ");  
        scanf("%d", &choice);  
        switch (choice) {  
            case 1: leftDistributive(); break;  
            case 2: rightDistributive(); break;  
            case 3: printf("Exiting program.\n"); break;  
            default: printf("Invalid choice. Please enter 1 to 3.\n");  
        }  
    } while (choice != 3);  
    return 0;  
}
```

```
void leftDistributive(void) {  
    int m,n,p,i,j,k,verified=1;  
    do { printf("Enter m: "); scanf("%d",&m); if(m<=0) printf("m must be positive.\n"); } while(m<=0);  
    do { printf("Enter n: "); scanf("%d",&n); if(n<=0) printf("n must be positive.\n"); } while(n<=0);  
    do { printf("Enter p: "); scanf("%d",&p); if(p<=0) printf("p must be positive.\n"); } while(p<=0);  
    int A[m][n],B[n][p],C[n][p],BC[n][p],left[m][p],AB[m][p],AC[m][p],right[m][p];  
    printf("Enter elements of matrix A (%d x %d):\n",m,n);  
    for(i=0;i<m;i++) for(j=0;j<n;j++) scanf("%d",&A[i][j]);  
    printf("Enter elements of matrix B (%d x %d):\n",n,p);  
    for(i=0;i<n;i++) for(j=0;j<p;j++) scanf("%d",&B[i][j]);  
    printf("Enter elements of matrix C (%d x %d):\n",n,p);  
    for(i=0;i<n;i++) for(j=0;j<p;j++) scanf("%d",&C[i][j]);  
    for(i=0;i<n;i++) for(j=0;j<p;j++) BC[i][j]=B[i][j]+C[i][j];  
    for(i=0;i<m;i++) for(j=0;j<p;j++) { left[i][j]=0; AB[i][j]=0; AC[i][j]=0; for(k=0;k<n;k++){ left[i][j]+=A[i][k]*BC[k][j];  
AB[i][j]+=A[i][k]*B[k][j]; AC[i][j]+=A[i][k]*C[k][j]; } right[i][j]=AB[i][j]+AC[i][j]; }  
    printf("\nA x (B+C):\n"); for(i=0;i<m;i++){ for(j=0;j<p;j++) printf("%d ",left[i][j]); printf("\n"); }  
    printf("\n(A x B) + (A x C):\n"); for(i=0;i<m;i++){ for(j=0;j<p;j++){ printf("%d ",right[i][j]); if(left[i][j]!=right[i][j]) verified=0; }  
printf("\n"); }  
    printf(verified ? "Left distributive property is verified.\n" : "Left distributive property is not verified.\n");  
}
```

```
void rightDistributive(void) {  
    int m,n,p,i,j,k,verified=1;  
    do { printf("Enter m: "); scanf("%d",&m); if(m<=0) printf("m must be positive.\n"); } while(m<=0);  
    do { printf("Enter n: "); scanf("%d",&n); if(n<=0) printf("n must be positive.\n"); } while(n<=0);  
    do { printf("Enter p: "); scanf("%d",&p); if(p<=0) printf("p must be positive.\n"); } while(p<=0);  
    int A[m][n],B[m][n],C[n][p],ABsum[m][n],left[m][p],AC[m][p],BC[m][p],right[m][p];  
    printf("Enter elements of matrix A (%d x %d):\n",m,n); for(i=0;i<m;i++) for(j=0;j<n;j++) scanf("%d",&A[i][j]);  
    printf("Enter elements of matrix B (%d x %d):\n",m,n); for(i=0;i<m;i++) for(j=0;j<n;j++) scanf("%d",&B[i][j]);  
    printf("Enter elements of matrix C (%d x %d):\n",n,p); for(i=0;i<n;i++) for(j=0;j<p;j++) scanf("%d",&C[i][j]);  
    for(i=0;i<m;i++) for(j=0;j<n;j++) ABsum[i][j]=A[i][j]+B[i][j];  
    for(i=0;i<m;i++) for(j=0;j<p;j++){ left[i][j]=0; AC[i][j]=0; BC[i][j]=0; for(k=0;k<n;k++){ left[i][j]+=ABsum[i][k]*C[k][j];  
AC[i][j]+=A[i][k]*C[k][j]; BC[i][j]+=B[i][k]*C[k][j]; } right[i][j]=AC[i][j]+BC[i][j]; }  
    printf("\n(A+B) x C:\n"); for(i=0;i<m;i++){ for(j=0;j<p;j++) printf("%d ",left[i][j]); printf("\n"); }  
    printf("\n(A x C) + (B x C):\n"); for(i=0;i<m;i++){ for(j=0;j<p;j++){ printf("%d ",right[i][j]); if(left[i][j]!=right[i][j]) verified=0; }  
printf("\n"); }  
    printf(verified ? "Right distributive property is verified.\n" : "Right distributive property is not verified.\n");  
}
```

Screenshot Spaces

Screenshot 1 - Successful Compilation

```
(kali㉿kali)-[~/Desktop]
$ vi hdoc34.c

(kali㉿kali)-[~/Desktop]
$ gcc hdoc34.c -o hdoc34 -lm

(kali㉿kali)-[~/Desktop]
$ ./hdoc34
```

Screenshot 2 - Choice 1: Left Distributive Property Output

```
— MATRIX DISTRIBUTIVE PROPERTY MENU —
1. Verify left distributive property
2. Verify right distributive property
3. Exit
Enter your choice: 1
Enter m: 2
Enter n: 2
Enter p: 2
Enter elements of matrix A (2 x 2):
1 2
3 4
Enter elements of matrix B (2 x 2):
5 6
7 8
Enter elements of matrix C (2 x 2):
1 1
1 1

A x (B+C):
22 25
50 57

(A x B) + (A x C):
22 25
50 57
Left distributive property is verified.
```

```

— MATRIX DISTRIBUTIVE PROPERTY MENU —
1. Verify left distributive property
2. Verify right distributive property
3. Exit
Enter your choice: 1
Enter m: 1
Enter n: 2
Enter p: 1
Enter elements of matrix A (1 x 2):
2 3
Enter elements of matrix B (2 x 1):
1
4
Enter elements of matrix C (2 x 1):
5
2
A x (B+C):
30
(A x B) + (A x C):
30
Left distributive property is verified.

```

```

— MATRIX DISTRIBUTIVE PROPERTY MENU —
1. Verify left distributive property
2. Verify right distributive property
3. Exit
Enter your choice: 1
Enter m: 0
m must be positive.
Enter m: 2
Enter n: 0
n must be positive.
Enter n: 2
Enter p: 2
Enter elements of matrix A (2 x 2):
1 2
3 4
Enter elements of matrix B (2 x 2):
5 6
7 8
Enter elements of matrix C (2 x 2):
1 1
1 1
A x (B+C):
22 25
50 57
(A x B) + (A x C):
22 25
50 57
Left distributive property is verified.

```

Screenshot 3 - Choice 2: Right Distributive Property Output

```
— MATRIX DISTRIBUTIVE PROPERTY MENU —
1. Verify left distributive property
2. Verify right distributive property
3. Exit
Enter your choice: 2
Enter m: 2
Enter n: 2
Enter p: 2
Enter elements of matrix A (2 x 2):
1 2
3 4
Enter elements of matrix B (2 x 2):
5 6
7 8
Enter elements of matrix C (2 x 2):
1 0
0 1
(A+B) x C:
6 8
10 12
(A x C) + (B x C):
6 8
10 12
Right distributive property is verified.
```

```
— MATRIX DISTRIBUTIVE PROPERTY MENU —
1. Verify left distributive property
2. Verify right distributive property
3. Exit
Enter your choice: 2
Enter m: 1
Enter n: 2
Enter p: 1
Enter elements of matrix A (1 x 2):
1 2
Enter elements of matrix B (1 x 2):
3 4
Enter elements of matrix C (2 x 1):
5
2
(A+B) x C:
32
(A x C) + (B x C):
32
Right distributive property is verified.
```

```
— MATRIX DISTRIBUTIVE PROPERTY MENU —
1. Verify left distributive property
2. Verify right distributive property
3. Exit
Enter your choice: 2
Enter m: 1
Enter n: 1
Enter p: -1
p must be positive.
Enter p: 2
Enter elements of matrix A (1 x 1):
2
Enter elements of matrix B (1 x 1):
3
Enter elements of matrix C (1 x 2):
4 5

(A+B) x C:
20 25

(A x C) + (B x C):
20 25
Right distributive property is verified.
```

Screenshot 4 - Invalid Input / Exit Output

```
— MATRIX DISTRIBUTIVE PROPERTY MENU —
1. Verify left distributive property
2. Verify right distributive property
3. Exit
Enter your choice: 9
Invalid choice. Please enter 1 to 3.
```

```
— MATRIX DISTRIBUTIVE PROPERTY MENU —
1. Verify left distributive property
2. Verify right distributive property
3. Exit
Enter your choice: 3
Exiting program.
```